

Key Objective:

Combine strong analytical skills in data with the practical ability to design and deploy AI agents that generate measurable business impact.

1. Understanding the “Tech 3 – Master” Profile

Advanced analytical reasoning

Technical autonomy

Strategic business thinking

Ability to propose leverage points

Capability to design end-to-end solutions

2. Data Analysis & Business Levers (Critical)**Advanced SQL**

CTEs

Window functions

Complex aggregations

Correlated subqueries

Query optimization

Dimensional modeling (Star Schema)

Business Metrics and Levers

Conversion funnels

Cohort analysis

Retention metrics

LTV (Lifetime Value)

CAC (Customer Acquisition Cost)

ROI (Return on Investment)

Price elasticity

3. AI Development – Agents

Core Knowledge

LLM fundamentals

RAG (Retrieval-Augmented Generation)

Tool integration

Orchestration

Memory strategies

Multi-agent systems

Observability and monitoring

Technical Foundations

Embeddings

Vectorization

Vector databases (FAISS, Pinecone)

Chunking strategies

LLM evaluation frameworks

Cost vs performance trade-offs

4. Generative AI Depth

Frontier models

Context windows

Hallucination causes

Alignment principles

Safety mechanisms

Temperature / top-p

Prompt chaining

Structured outputs

Ability to justify technical decisions

5. Large-Scale Data Systems (Desirable)

Partitioning strategies

Indexing techniques

Query execution plans

Distributed systems concepts

Spark fundamentals

ETL vs ELT

Data warehouse vs Data lake

6. 3-Week Intensive Study Plan

Week 1 – SQL & Business Analytics

Advanced SQL practice

Data modeling

Business metrics deep dive

Churn case study

Week 2 – Generative AI & Agents

Build an agent with tool calling

Implement a RAG pipeline

Evaluate model outputs

Measure inference cost

Week 3 – Architecture & Strategy

Design an end-to-end solution

Address scalability concerns

Simulate a technical interview

Prepare a project pitch

7. Interview Preparation

Be ready to answer:

How would you measure the impact of an AI agent?

How do you ensure reliability?

How do you reduce hallucinations?

How do you scale an LLM-based system?

How do you justify cost vs benefit?